

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques

using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I **T.GANESH(192425246)**(Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

T.GANESH

Signature of the student:

1. Smart Campus – 192.168.100.0/24

Given block 192.168.100.0/24 → 256 addresses
(192.168.100.0 – 192.168.100.255)

We allocate the largest subnet first (VLSM – Variable Length Subnet Mask) and place them consecutively from the start of the block.

Step-by-step allocation

Order	Department	Required Mask	Block Size	Starting Address	Ending Address
1	School of Computing	/25	128	192.168.100.0	192.168.100.127
2	School of Electronics	/26	64	192.168.100.128	192.168.100.191
3	School of Mechanical Engineering	/27	32	192.168.100.192	192.168.100.223

Detailed results

Department	Network Address	Broadcast Address	Valid Host Range	Usable Hosts	Binary View of Last Octet
School of Computing (/25)	192.168.100.0	192.168.100.127	192.168.100.1 – 192.168.100.126	126	0xxxxxxx
School of Electronics (/26)	192.168.100.128	192.168.100.191	192.168.100.129 – 192.168.100.190	62	10xxxxxx
School of Mechanical Engg (/27)	192.168.100.192	192.168.100.223	192.168.100.193 – 192.168.100.222	30	110xxxxx

Address-space diagram (ASCII)

text

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192.168.100.0/24 (full block)

└── 192.168.100.0/25 School of Computing (128 addresses)

| └── Hosts: .1 – .126
 |── 192.168.100.128/26 School of Electronics (64
 addresses)
 | └── Hosts: .129 – .190
 |── 192.168.100.192/27 School of Mechanical Engg (32
 addresses)
 | └── Hosts: .193 – .222
 └── 192.168.100.224 – .255 UNUSED (32
 addresses left)

Remaining unused range 192.168.100.224 –
 192.168.100.255 (can still be divided into two /28s or one /27
 later).

2. Multinational Software Company – 10.10.0.0/16

Given block 10.10.0.0/16 → 65 536 addresses

Again we use VLSM and allocate from the lowest address upward.

Department	Mask	Size	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
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Software Development	/24	256	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	/25	128	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	/26	64	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	/27	32	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

Address-space diagram

text

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10.10.0.0/16

└─ 10.10.0.0/24 Software Development

└─ 10.10.1.0/25 Quality Assurance

└─ 10.10.1.128/26 Human Resources

└─ 10.10.1.192/27 Executive Management

└─ 10.10.1.224 – 10.10.255.255 UNUSED (huge remaining space)

Remaining unused range 10.10.1.224 – 10.10.255.255
(approximately 65 000+ addresses still free).

3. Smart Hospital – IPv6 Migration

Given address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed notation (RFC 5952 rules)

- Leading zeros in each hextet are removed.
- Longest consecutive sequence of zero hextets is replaced by :: (only once). → **2001:db8:abcd::/64**

2. Full expanded form

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network prefix (/64) The first 64 bits are the network portion: 2001:0db8:abcd:0000::/64 (or compressed: 2001:db8:abcd::/64)

4 & 5. Total host addresses A /64 prefix leaves 64 bits for the Interface ID (host portion). Total addresses = 2^{64} = **18 446 744 073 709 551 616** (In IPv6 almost all of these can be used by hosts; there is no “network” or “broadcast” reservation like IPv4.)

Visual representation

text

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2001:0db8:abcd:0000 : 0000:0000:0000:0000

|<----- 64-bit Network Prefix ----->|<----- 64-bit Interface ID ----->|

4. ISP Core Router – Packet Forwarding

Routing table

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192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Packet destination: 192.168.2.45

Step-by-step decision

1. The router performs a **longest-prefix-match** lookup.
2. 192.168.2.45 falls inside the range 192.168.2.0 – 192.168.2.255.
3. Therefore the matching entry is **192.168.2.0/24**.
4. The packet is forwarded out of the interface (or to the next-hop IP) that is associated with that route.
5. If the destination had **not** matched any of the three entries, the router would use the **default route 0.0.0.0/0**.
- 6.

Simple topology diagram

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(Customers A) (Customers B) (Customers C)

5. Two Office Departments (Layer-3 Router)

- Administration LAN $\rightarrow 192.168.10.0/26$
- Finance LAN $\rightarrow 192.168.10.64/26$

LAN	Network Address	Broadcast Address	Valid Host Range	Usable Hosts	Default Gateway
Administration	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62	62	192.168.10.1
Finance	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126	62	192.168.10.65

Suggested IP addresses for three computers in each LAN

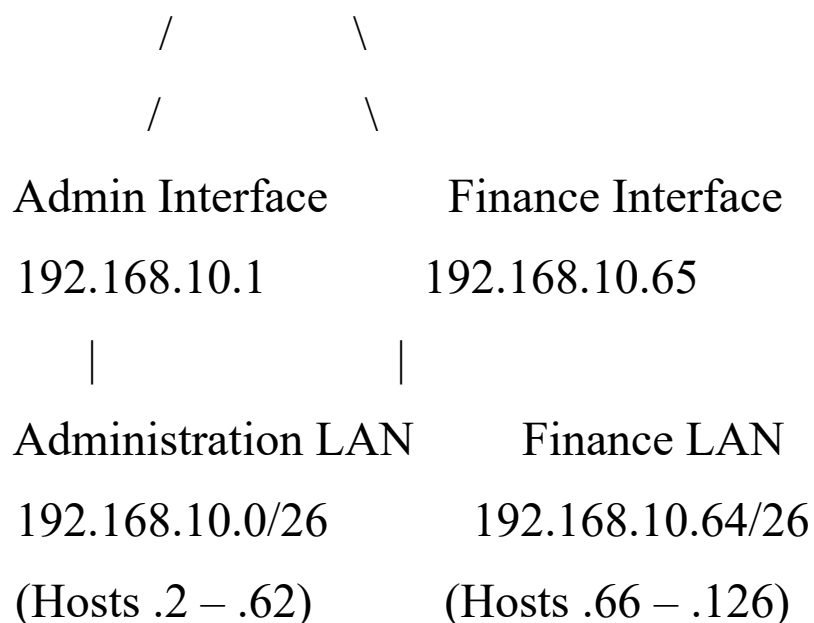
- **Administration LAN** PC1 → 192.168.10.10 PC2 → 192.168.10.20 PC3 → 192.168.10.30 (Gateway = 192.168.10.1)
- **Finance LAN** PC1 → 192.168.10.70 PC2 → 192.168.10.80 PC3 → 192.168.10.90 (Gateway = 192.168.10.65)

Network topology diagram

text

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[Layer-3 Router]



RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand